

**CAT I: 3<sup>rd</sup> YEAR SECOND SEMESTER, BSc MECHATRONIC  
ENGINEERING**

**EMT 3205 MEASUREMENT AND INSTRUMENTATION**

**Q1.** Name and briefly explain the two main types of errors that arise during the measurement process. [4 marks]

**Q2.** Explain why active instruments generally provide better measurement resolution than passive instruments. [2marks]

**Q3.** Explain the difference between accuracy and precision in measurement systems. Why is it important to distinguish between these two concepts in engineering practice? [4 marks]

**Q4.** Explain why systematic errors are considered more problematic than random errors in measurement systems, especially in applications such as quality control. [3 marks]

**Q5.** State and explain three characteristics that distinguish digital instruments from analog instruments. [3marks]

**Q6.** List three examples of resistive transducers and state what physical quantity each measures. [6 marks]

Maxwell's -  $\frac{R_1 R_3}{R_2 R_4}$   
Hay's -  $\frac{R_1 R_3}{R_2 R_4}$   
Schering -  $\frac{R_1 R_3}{R_2 R_4}$   
Klein -  $\frac{R_1 R_3}{R_2 R_4}$

$$L_1 = \frac{R_3}{R_4} \times L_2$$
$$R_1 = \frac{R_3}{R_4} \times R_2$$

**Q5.** A spring balance is being evaluated for use in a pharmaceutical production facility. The device was calibrated at 22°C with the following performance:

- Load of 50 kg → Deflection of 20 mm
- Load of 100 kg → Deflection of 40 mm
- Load of 150 kg → Deflection of 60 mm

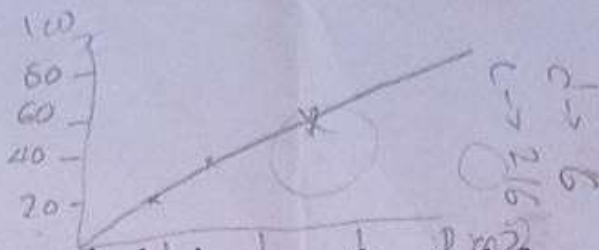
During winter operation, when the facility temperature drops to 5°C, repeat measurements of a 100 kg standard weight produce the following deflections:

- Measurement 1: 39.2 mm
- Measurement 2: 39.1 mm
- Measurement 3: 39.3 mm
- Measurement 4: 39.0 mm

i) Calculate the calibration sensitivity of the spring balance at 22°C. [2 marks]

ii) Determine whether zero drift, sensitivity drift, or both are present at 5°C. Show your calculations. [3 marks]

iii) For pharmaceutical production where accuracy is critical, explain what measurement error types have occurred and recommend what corrective action should be taken. [3 marks]



$L \rightarrow 216 \text{ mm}$   
 $P_{max} \rightarrow 29 \text{ N}$   
 $C_{hr} \rightarrow 6$

$T_{org} \rightarrow 1023$   
 $OD \rightarrow 16 \text{ mm}$   
 $ID \rightarrow 19 \text{ mm}$

$$\begin{aligned}
 2.5 \text{ kg} &= 1 \text{ mm} \\
 1 \text{ kg} &= 0.4 \text{ mm} \\
 1 \times \frac{1}{2.5} &= 0.4 \\
 2.554 &= 1 \text{ mm} \\
 1 &= 0.3915 \text{ mm}
 \end{aligned}$$

39.15

systematic error

412

824

2

section view

material: Al 6061-T6